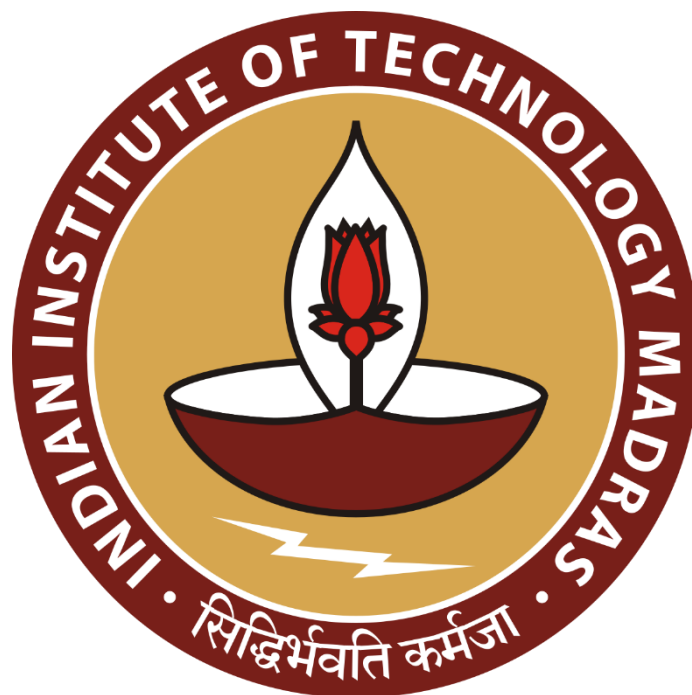


Nutritional Analysis and Strategic Menu Engineering for McDonald's India

A Proposal report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project Title “**Nutritional Analysis and Strategic Menu Engineering for McDonald's India**”. I am conducting this research using publicly available data to fulfill the requirements of the BDM Capstone project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through secondary **sources (Kaggle)** and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of Candidate:

Name: Tushar Pillay

1 Executive Summary and Title

Title: Nutritional Analysis and Strategic Menu Engineering for McDonald's India

This project performs a data-driven evaluation of the nutritional landscape of McDonald's India to address the growing consumer shift toward health-consciousness. As a leading B2C organization in the Quick Service Restaurant (QSR) sector, McDonald's faces a significant business challenge: balancing the high-velocity sales of traditional "Value" items with the increasing demand for nutritional transparency and better dietary choices.

The primary business issues identified involve a **"Health-Profit Paradox,"** where high-calorie and high-sodium items may create a brand risk among wellness-oriented demographics. Specifically, the menu exhibits "Hidden Nutritional Variance," where large serving sizes often contribute excessive empty calories without a proportional increase in protein a metric defined here as **"Satiety Efficiency."** Furthermore, the absence of a data-driven "Healthy Bundle" strategy prevents the organization from capturing the emerging market segment seeking meal combinations strictly under a 600-kCal threshold.

These issues will be addressed by analyzing the menu dataset through **Descriptive Statistics** to establish caloric baselines and **Correlation Analysis** to examine the relationship between `Per Serve Size` and `Sodium` content.

Additionally, **Combinatorial Analysis** will be used to design optimized meal bundles, while **Index Modeling** will create a "Nutrient-Density Score" for every item. The expected outcome is a strategic framework for menu optimization and "Nutritionally Bankrupt" item pruning, which helps increase long-term brand loyalty and profitability.

2 Organization Background

McDonald's India is a dominant player in the Indian Quick Service Restaurant (QSR) industry, currently operating a vast network of **over 800 restaurants** across the country. The brand's operations are managed through two major developmental licensees: **Westlife Foodworld Ltd.**, overseeing over 450 outlets in West and South India, and **Connaught Plaza Restaurants Pvt. Ltd. (CPRL)**, managing approximately 350 outlets in the North and East regions.

Since its inception in 1996, McDonald's has pioneered a "Glocal" strategy, successfully adapting its global business model to suit Indian cultural and religious sensitivities. This led to the creation of a unique **"No Beef, No Pork"** supply chain and the introduction of localized icons like the McAloo Tikki and Maharaja Mac. With a presence in over 75 cities, the organization maintains a rigorous, temperature-controlled "Cold Chain" infrastructure to serve millions of customers daily. This commitment to local adaptation and operational excellence makes McDonald's India a premier subject for a Business Data Management (BDM) case study.

3 Problem Statement

Problem Statement 1: Menu Pruning (The Health-Profit Paradox)

There is a gap in identifying "Nutritionally Bankrupt" items that contribute high operational costs but offer low nutritional value. This study aims to identify items with high Sodium (mg) and Total Fat (g) but low Protein (g) to recommend their removal from the menu.

Problem Statement 2: Optimization of "Healthy Bundles" (Product Bundling)

Current meal combinations often exceed daily recommended caloric limits, leading to customer "guilt" and brand risk. I intend to use **Combinatorial Analysis** on the Energy (kCal) column to design optimized meal bundles that remain strictly under a 600-kCal threshold.

Problem Statement 3: Satiety Efficiency (Portion Optimization)

Increasing Per Serve Size often adds "empty calories" without a proportional increase in satiating nutrients like Protein. This research will quantify the **Nutrient-**

to-Weight ratio to identify which large portions fail to provide efficient nutritional value to the consumer.

Problem Statement 4: Nutrient-Density Index (Strategic Positioning)

The organization lacks a single metric to communicate "Health Value" to consumers across different **Menu Categories**. I will establish a **Nutrient-Density Index** by correlating **Energy** against **Protein** and **Sugars** to rank the entire menu for better strategic positioning.

4 Background of the Problem

The nutritional challenge at McDonald's India arises from a misalignment between historical "Value-based" menu engineering and the modern Indian consumer's shift toward wellness. This problem has surfaced as a significant **gap in knowledge** regarding the actual nutritional efficiency of localized menu items versus their perceived value.

Major Causes of the Problem:

- **Internal Factors:** The primary internal driver is the "**Health-Profit Paradox**." To maintain low price points, the organization may rely on high-sodium preservatives and fats to enhance flavour and shelf-life across its **800+ outlets**. Currently, there is a lack of a unified **Nutrient-Density Index** to audit these items, leading to "Hidden Nutritional Variance" where certain "Vegetarian" or "Breakfast" categories inadvertently carry higher sodium loads than standard burgers.
- **External Factors:** Externally, the **FSSAI (Food Safety and Standards Authority of India)** is implementing stricter "Front-of-Pack" labeling norms. Simultaneously, a more informed middle-class demographic is scrutinizing Total Sugars and Trans Fats. Failure to provide "Healthy Bundles" under a **600-kCal threshold** creates a competitive disadvantage against "healthy-label" rivals.

These factors combined have created a critical need for data-driven menu pruning and portion optimization to mitigate brand risk and ensure long-term satiety efficiency for the customer.

5 Problem Solving Approach

To address the identified gaps in McDonald's India's menu strategy, I will implement a multi-stage data science and business analytical approach. This structured

methodology moves from raw data processing to high-level strategic modeling, ensuring each business problem is mitigated with empirical evidence.

5.1 Data Source & Proof of Originality

To ensure the transparency and reliability of this research, the primary data is sourced from a verified secondary repository. In accordance with the BDM guidelines for the secondary data path, the link to the raw dataset used for this analysis is provided below:

- **Repository:** Kaggle
- **Dataset Title:** McDonald's India : Menu Nutrition Dataset
- **URL:** <https://www.kaggle.com/datasets/deepcontractor/mcdonalds-india-menu-nutrition-facts>

Step 1: Data Pre-processing and Standardization (Power Query)

The first step involves importing the secondary dataset() into Excel using **Power Query**. I will perform **Data Transformation** on the `Per Serve Size` column by using the **"Split Column by Delimiter"** feature to remove text units (e.g., "g" or "ml"), converting it into a purely numerical variable. To handle missing values in the `Sodium (mg)` and `Added Sugars (g)` columns, I will use **Mean Imputation** calculated through **Pivot Tables** to ensure the values are representative of their respective `Menu Category`, thereby maintaining data integrity.

Step 2: Descriptive Analysis & Outlier Detection (Data Analysis Toolpak)

To solve **Problem 1 (Menu Pruning)**, I will generate **Descriptive Statistics** (Mean, Median, Mode, and Standard Deviation) using the **Excel Data Analysis Toolpak**. I will implement **Conditional Formatting** and **Z-Score Analysis** to identify "Nutritionally Bankrupt" outliers—items residing in the top 10th percentile for `Sodium` and `Total Fat` but falling below the 25th percentile for `Protein`. This visual identification serves as the primary basis for identifying items that damage the brand's health-conscious positioning.

Step 3: Correlation & Variance Analysis (Bivariate Modeling)

To address **Problem 3 (Satiety Efficiency)** and **Problem 4 (Hidden Sodium Risk)**, I will utilize Excel's `CORREL` function to calculate the **Pearson Correlation Coefficient** between `Per Serve Size` and `Protein`. This will help determine the **Nutrient-to-Weight Ratio**, identifying if increasing portion sizes adds genuine value or merely "Empty Calories." Simultaneously, I will use **ANOVA: Single**

Factor from the Toolpak to identify statistically significant **Sodium** variances across different **Menu Categories**, uncovering "Hidden Salt" spikes that deviate from consumer expectations.

Step 4: Combinatorial Modeling for Healthy Bundles (Excel Solver)

To mitigate **Problem 2 (600-kCal Bundling)**, I will implement a **Combinatorial Optimization** model using **Excel Solver**. By setting the objective to maximize variety while keeping the sum of **Energy (kCal)** across three distinct categories—Main Course, Sides, and Beverages—under a constraint of 600 kCal, I will identify unique meal permutations. This provides a data-backed foundation for the "Smart Combo" recommendation engine.

Step 5: Metric Development (Nutrient-Density Index)

Finally, I will develop a weighted **Nutrient-Density Index (NDI)** using Excel formulas. This model will assign a custom score by rewarding **Protein** content while penalizing **Total Sugars** and **Sodium**. This single metric will allow the organization to rank the entire menu on a unified "Health-Value" scale, facilitating strategic product positioning and ensuring compliance with **FSSAI labeling norms**.

6 Expected Timeline

6.1 Work Breakdown Structure (WBS)

The project is organized into four primary functional phases, focusing on data integrity and strategic menu modeling.

- **Phase 1: Project Initiation & Data Acquisition**
 - 1.1 Selection of McDonald's India Dataset (Kaggle Repository).
 - 1.2 Data Repository verification and Proof of Originality documentation.
 - 1.3 Proposal drafting and submission for "Accept/Reject" status.
- **Phase 2: Data Pre-processing (Excel Power Query)**
 - 2.1 Data Cleaning: Stripping text units from Per Serve Size.
 - 2.2 Handling Missing Values: Mean Imputation for Sodium and Sugar.
 - 2.3 Metadata Creation: Defining variables and data types.

- **Phase 3: Exploratory & Statistical Analysis (Excel)**
 - 3.1 Descriptive Statistics: Calculating Mean/Median for all 7 menu categories.
 - 3.2 Outlier Detection: Z-Score analysis to identify "Nutritionally Bankrupt" items.
 - 3.3 Correlation Analysis: Determining the Nutrient-to-Weight ratio.
- **Phase 4: Strategic Modeling & Final Reporting**
 - 4.1 Combinatorial Optimization: Using Excel Solver for 600-kCal bundles.
 - 4.2 Index Development: Creating the weighted Nutrient-Density Index (NDI).
 - 4.3 Final Report Compilation: Drafting the 20-page document and Plagiarism check.
 - 4.4 Viva Voce Preparation: Practicing the live data demonstration.

7 Expected Outcome (Briefly explain in 150-200 words)

The primary outcome of this project is a data-backed **Strategic Menu Roadmap** for McDonald's India that aligns product offerings with increasing consumer health-consciousness. By the end of this study, the organization will achieve the following:

- 7.1 **Menu Optimization & Pruning:** A definitive "Red-List" of **Nutritionally Bankrupt** items will be identified—those with high sodium/fat and low protein—allowing the company to prune low-value products and reduce operational complexity in its **800+ outlets**.
- 7.2 **Optimized Product Bundling:** The development of a "**Healthy Bundle Matrix**" will provide pre-calculated meal combinations (Main + Side + Drink) that strictly adhere to a **600-kCal threshold**, creating a new revenue stream for health-aware customers.
- 7.3 **Satiety & Portion Efficiency:** A better understanding of the **Nutrient-to-Weight ratio** will identify which large serving sizes contribute "empty calories" versus genuine satiety, enabling better portion-size engineering.

- 7.4 **7.4 Strategic Positioning:** The implementation of a **Nutrient-Density Index (NDI)** will provide a single, transparent metric to rank the entire menu. This ensures better compliance with evolving **FSSAI labeling norms** and mitigates the risk of customer churn to "healthy-label" competitors.
- 7.5 Ultimately, these outcomes will help McDonald's India increase brand loyalty and long-term profitability by transitioning from a "Volume-based" to a "**Value-based**" nutritional strategy.

-----End of the report-----